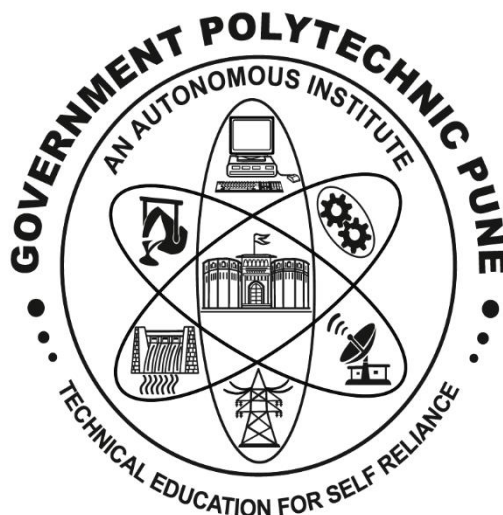


GOVERNMENT POLYTECHNIC PUNE

(An Autonomous Institute of Government of Maharashtra)



MICPROJECT REPORT ON “CAMPUS LOST & FOUND”

For the partial fulfilment of Diploma in Computer Engineering, GP Pune.

Guided By

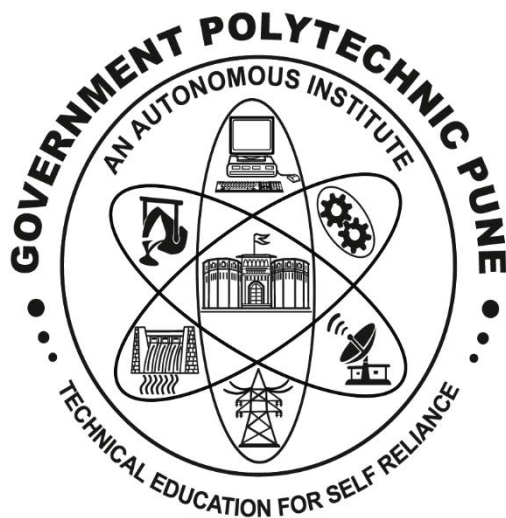
Mrs. Vaishali S. Pawar

Submitted By

Sr.No.	Name of the Student	Enrollment No.
1.	Nakshatra Kadam	2406212

GOVERNMENT POLYTECHNIC PUNE

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CERTIFICATE

This is to certify that the mini-microproject work entitled “**CAMPUS LOST & FOUND**” is a Bonafide work carried out by

Nakshatra Kadam 2406212

Of Class Third year in partial fulfilment of required for completion of course
Mobile Application Development in Computer Engineering from
Government Polytechnic Pune. The report has been approved as it satisfies the
academic requirements in respect of micro-project work prescribed for the
course.

Microproject Guide

Mrs. Vaishali S. Pawar

HOD

Mrs. J.R. Hange

Principal

Dr. R.K. Patil

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guide **Mrs. Vaishali S. Pawar** for providing us with valuable guidance, support and encouragement throughout the development of this microproject.

We are thankful to the **Department of Computer Engineering, Government Polytechnic Pune**, for providing us with the necessary facilities and resources required to complete this project.

We also thank our faculty members for their valuable suggestions and technical guidance during the development and testing of the application.

Finally, we would like to thank our friends and classmates for their support and feedback during the development of the **Campus Lost & Found** application.

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INTRODUCTION

The **Campus Lost & Found** is a Native Android application designed to help students and staff report and recover lost or found items within a university or college campus.

In educational institutions, students frequently lose personal belongings such as wallets, identity cards, keys, headphones, calculators, mobile phones and other valuable items. Traditional methods of finding these items usually depend on notice boards, messaging groups or manually contacting other students.

The proposed application provides a centralized digital platform where users can report lost or found items, browse available posts and search for specific items.

The application also provides **campus-based isolation**, where users join a campus using a unique campus code. Items displayed in the feed are restricted to the currently selected campus.

The application is developed using **Java and XML in Android Studio**, with **SQLite** used for local data storage.

OBJECTIVES

1. To provide a centralized platform for reporting lost and found items.
2. To allow users to register and securely log in to the application.
3. To separate items according to different university or campus communities.
4. To allow users to upload details and photographs of lost or found items.
5. To provide separate Lost and Found feeds for easy browsing.
6. To provide search and filtering functionality for finding relevant items.
7. To allow users to contact the person who posted an item.
8. To allow users to manage and resolve their own uploaded items.
9. To store application data locally using SQLite.
10. To demonstrate Android UI, activities, intents, permissions, adapters and database concepts.

LITERATURE REVIEW / THEORETICAL BACKGROUND

3.1 Android Application Development

Android is an open-source mobile operating system used for developing applications for smartphones and other devices. Android applications can be developed using Android Studio with Java or Kotlin. In this project, Java is used for application logic and XML is used for designing the user interface.

3.2 Activities and Intents

An Activity represents a screen of an Android application. The project uses multiple activities for login, campus selection, Lost and Found feeds, item reporting, search and My Uploads.

Intent is used to navigate between activities and perform actions such as opening the phone dialer.

3.3 Android UI Components

The application uses Android UI components such as:

- TextView
- EditText
- Button
- ImageView
- ListView
- LinearLayout
- ConstraintLayout
- ScrollView
- Dialogs
- Toast messages

These components are used to create an interactive and user-friendly interface.

3.4 ListView and Adapter

ListView is used to display multiple Lost and Found item records.

An Adapter connects the data with the ListView and creates the layout for each item. The project uses a custom ItemAdapter to display the item image, title, description, location, date, uploader information and contact button.

3.5 SQLite Database

SQLite is a lightweight relational database available in Android. The project uses SQLiteOpenHelper to create and manage the local database.

The database contains three main tables:

- users – stores user information.
- campuses – stores campus information.
- items – stores Lost and Found item details.

SQLite operations such as insertion, querying and deletion are used to manage application data.

3.6 SharedPreferences

SharedPreferences provides simple persistent storage for small amounts of application data.

The project uses it to maintain the user's login session and store the currently selected campus.

3.7 Android Permissions

Android uses a permission system to protect sensitive device resources. The application requests Camera permission when the user wants to capture an image of a Lost or Found item.

Runtime permission handling ensures that permission is requested when the feature is actually used.

PROJECT DESCRIPTION / METHODOLOGY

Campus Lost & Found is a Native Android application that allows students and staff to report, browse, search and resolve Lost and Found items within their campus.

The application uses **Java and XML** for development and **SQLite** for local data storage.

Basic Workflow

Login/Register → Join Campus → View Items → Report/Search Item → Contact User → Resolve Item

4.2 Main Modules

Authentication

- User registration and login.
- Stores user session using SharedPreferences.

Campus Management

- Users join a campus using a unique campus code.
- Items are displayed according to the selected campus.

Lost & Found

- Separate feeds for Lost and Found items.
- Displays item image, description, location, date and contact details.

Report Item

- Users can add Lost or Found items.
- Supports image selection from Gallery and Camera.

Search

- Searches items using title, location and description.
- Provides Lost, Found and All filters.

My Uploads

- Displays items uploaded by the current user.
- Allows users to resolve and remove their own posts.

IMPLEMENTATION / EXPERIMENTATION

5Development Environment

The application is developed using **Android Studio** with **Java** as the programming language and **XML** for designing the user interface. SQLite is used as the local database. The application has a minimum SDK of API 26 and uses API 35 for compilation and targeting.

Application Structure

The application is divided into different activities and supporting Java classes. The major activities include MainActivity, LoginActivity, JoinCampusActivity, FoundActivity, LostActivity, AddFoundItemActivity, AddLostItemActivity, SearchActivity and MyUploadsActivity.

Supporting classes such as DatabaseHelper, ItemAdapter, ItemModel, SessionManager and CampusManager handle database operations, item display, data management, login sessions and campus selection.

Application Flow

1. The application starts by checking the user's login session.
2. New users can register, while existing users can log in.
3. After login, the user joins a campus using a valid campus code.
4. The selected campus is stored for maintaining campus-specific data.
5. Users can view separate Lost and Found item feeds.
6. Items are retrieved from SQLite and displayed using a custom ListView adapter.
7. Users can report Lost or Found items with details and photographs.
8. The Search screen allows users to find items using title, location or description.
9. The Direct Call option opens the phone dialer using an Android Intent.
10. Users can manage and resolve their uploaded items through My Uploads.

Database Implementation

SQLite is used to store application data locally. The database contains records for users, campuses and Lost or Found items.

SQLiteOpenHelper is used to create and manage the database. Operations such as inserting new records, retrieving records using queries and deleting resolved items are performed through SQLite.

Transactions are used while saving important item records to maintain database consistency.

Image and Permission Handling

The application allows users to select item photographs from the Gallery or capture them using the device Camera.

Camera access requires runtime permission. The application requests this permission when the camera feature is used and proceeds with image capture only after permission is granted.

CODE

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#1E293B">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="Campus Lost & Found"
        android:textColor="@android:color/white"
        android:textSize="26sp"
        android:textStyle="bold" />

</FrameLayout>
```

MainActivity.java:

```
package com.example.lostandfound;

import android.content.Intent;
import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        Intent intent;
        if (!SessionManager.isLoggedIn(this)) {
            intent = new Intent(MainActivity.this, LoginActivity.class);
        } else if (!CampusManager.hasJoinedCampus(this)) {
            intent = new Intent(MainActivity.this, JoinCampusActivity.class);
        } else {
```

```

        intent = new Intent(MainActivity.this, FoundActivity.class);
    }
    startActivity(intent);
    finish();
}
}

```

Activity_login.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ImageView
        android:id="@+id/backgroundImage"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:src="@drawable/campus_bg"
        android:scaleType="centerCrop" />

    <View
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:background="#66000000" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="24dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Campus Lost and Found"
            android:textColor="@android:color/white"
            android:textSize="28sp"
            android:textStyle="bold"
            android:layout_marginBottom="24dp" />

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"

```

```
android:orientation="horizontal">
```

```
<Button
    android:id="@+id/btnLoginTab"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:layout_marginEnd="6dp"
    android:text="LOGIN" />
```

```
<Button
    android:id="@+id/btnRegisterTab"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:layout_marginStart="6dp"
    android:text="REGISTER" />
```

```
</LinearLayout>
```

```
<LinearLayout
    android:id="@+id/loginForm"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:layout_marginTop="16dp">
```

```
<EditText
    android:id="@+id/loginEmail"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Email"
    android:inputType="textEmailAddress"
    android:backgroundTint="@android:color/white"
    android:textColor="@android:color/white"
    android:layout_marginTop="12dp"
    android:textColorHint="#DDDDDD" />
```

```
<EditText
    android:id="@+id/loginPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:inputType="textPassword"
    android:backgroundTint="@android:color/white"
    android:textColor="@android:color/white"
```

```
    android:layout_marginTop="12dp"
    android:textColorHint="#DDDDDD" />
```

```
<Button
    android:id="@+id/loginButton"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login"
    android:layout_marginTop="34dp" />
```

```
</LinearLayout>
```

```
<LinearLayout
    android:id="@+id/registerForm"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:layout_marginTop="16dp"
    android:visibility="gone">
```

```
<EditText
    android:id="@+id/registerName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Full Name"
    android:inputType="textPersonName"
    android:backgroundTint="@android:color/white"
    android:textColor="@android:color/white"
    android:layout_marginTop="12dp"
    android:textColorHint="#DDDDDD" />
```

```
<EditText
    android:id="@+id/registerEmail"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Email"
    android:inputType="textEmailAddress"
    android:backgroundTint="@android:color/white"
    android:textColor="@android:color/white"
    android:layout_marginTop="12dp"
    android:textColorHint="#DDDDDD" />
```

```
<EditText
    android:id="@+id/registerPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```
        android:hint="Password"
        android:inputType="textPassword"
        android:backgroundTint="@android:color/white"
        android:textColor="@android:color/white"
        android:layout_marginTop="12dp"
        android:textColorHint="#DDDDDD" />
```

```
<EditText
    android:id="@+id/registerConfirmPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Confirm Password"
    android:inputType="textPassword"
    android:backgroundTint="@android:color/white"
    android:textColor="@android:color/white"
    android:layout_marginTop="12dp"
    android:textColorHint="#DDDDDD" />
```

```
<Button
    android:id="@+id/registerButton"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Register"
    android:layout_marginTop="34dp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
</FrameLayout>
```

LoginActivity.java:

```
package com.example.lostandfound;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.LinearLayout;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class LoginActivity extends AppCompatActivity {
```

```

Button btnLoginTab, btnRegisterTab, loginButton, registerButton;
LinearLayout loginForm, registerForm;
EditText loginEmail, loginPassword;
EditText registerName, registerEmail, registerPassword, registerConfirmPassword;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_login);

    btnLoginTab = findViewById(R.id.btnLoginTab);
    btnRegisterTab = findViewById(R.id.btnRegisterTab);
    loginButton = findViewById(R.id.loginButton);
    registerButton = findViewById(R.id.registerButton);

    loginForm = findViewById(R.id.loginForm);
    registerForm = findViewById(R.id.registerForm);

    loginEmail = findViewById(R.id.loginEmail);
    loginPassword = findViewById(R.id.loginPassword);

    registerName = findViewById(R.id.registerName);
    registerEmail = findViewById(R.id.registerEmail);
    registerPassword = findViewById(R.id.registerPassword);
    registerConfirmPassword = findViewById(R.id.registerConfirmPassword);

    btnLoginTab.setOnClickListener(v -> {
        loginForm.setVisibility(View.VISIBLE);
        registerForm.setVisibility(View.GONE);
    });

    btnRegisterTab.setOnClickListener(v -> {
        loginForm.setVisibility(View.GONE);
        registerForm.setVisibility(View.VISIBLE);
    });

    loginButton.setOnClickListener(v -> {
        String email = loginEmail.getText().toString().trim();
        String password = loginPassword.getText().toString();

        if (email.isEmpty() || password.isEmpty()) {
            Toast.makeText(this, "Please enter email and password",
Toast.LENGTH_SHORT).show();
            return;
        }
    }

```

```

DatabaseHelper dbHelper = DatabaseHelper.getInstance(this);
boolean valid = dbHelper.validateUser(email, password);

if (valid || (email.equalsIgnoreCase("test@test.com") && password.equals("test"))) {
    String userName = dbHelper.getUserNameByEmail(email);
    if (userName.isEmpty()) {
        userName = "Test User";
    }
    SessionManager.setLoggedIn(this, true, email, userName);

    Intent intent;
    if (CampusManager.hasJoinedCampus(this)) {
        intent = new Intent(LoginActivity.this, FoundActivity.class);
    } else {
        intent = new Intent(LoginActivity.this, JoinCampusActivity.class);
    }
    startActivity(intent);
    finish();
} else {
    Toast.makeText(LoginActivity.this, "Invalid email or password",
Toast.LENGTH_SHORT).show();
}
});

registerButton.setOnClickListener(v -> {
    String name = registerName.getText().toString().trim();
    String email = registerEmail.getText().toString().trim();
    String password = registerPassword.getText().toString();
    String confirmPassword = registerConfirmPassword.getText().toString();

    if (name.isEmpty() || email.isEmpty() || password.isEmpty()) {
        Toast.makeText(this, "Please fill in all required fields",
Toast.LENGTH_SHORT).show();
        return;
    }

    if (!password.equals(confirmPassword)) {
        registerConfirmPassword.setError("Passwords do not match");
        registerConfirmPassword.requestFocus();
        return;
    }

    DatabaseHelper dbHelper = DatabaseHelper.getInstance(this);
    long id = dbHelper.registerUser(name, email, password, "", "");
    if (id != -1) {

```

```

        SessionManager.setLoggedIn(this, true, email, name);
        Toast.makeText(this, "Registration successful!", Toast.LENGTH_SHORT).show();

        Intent intent = new Intent(LoginActivity.this, JoinCampusActivity.class);
        startActivity(intent);
        finish();
    } else {
        Toast.makeText(this, "Email is already registered. Please login.",
            Toast.LENGTH_LONG).show();
    }
    });
}
}

```

activity_lost.xml

```

<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#EBE9E1">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:padding="16dp">

        <RelativeLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="12dp"
            android:layout_marginBottom="16dp">

            <ImageButton
                android:id="@+id/btnMenu"
                android:layout_width="40dp"
                android:layout_height="40dp"
                android:layout_alignParentStart="true"
                android:layout_centerVertical="true"
                android:background="?attr/selectableItemBackgroundBorderless"
                android:src="@drawable/ic_more_vert"
                app:tint="#1E293B"
                android:contentDescription="Menu" />

```

```
<TextView
    android:id="@+id/tvLostTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_centerInParent="true"
    android:text="Campus Lost"
    android:textColor="#1E293B"
    android:textSize="22sp"
    android:textStyle="bold" />
```

```
<ImageButton
    android:id="@+id/btnSearch"
    android:layout_width="40dp"
    android:layout_height="40dp"
    android:layout_alignParentEnd="true"
    android:layout_centerVertical="true"
    android:background="?attr/selectableItemBackgroundBorderless"
    android:src="@drawable/ic_search"
    app:tint="#1E293B"
    android:contentDescription="Search" />
```

```
</RelativeLayout>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center"
    android:layout_marginBottom="16dp">
```

```
<LinearLayout
    android:id="@+id/tabFound"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center_horizontal"
    android:paddingStart="16dp"
    android:paddingEnd="16dp"
    android:clickable="true"
    android:focusable="true">
```

```
<TextView
    android:id="@+id/tvTabFound"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Found"
```

```
    android:textColor="#8E8A82"
    android:textSize="18sp" />
```

```
<View
    android:id="@+id/indicatorFound"
    android:layout_width="32dp"
    android:layout_height="3dp"
    android:layout_marginTop="6dp"
    android:background="@drawable/bg_tab_indicator"
    android:visibility="invisible" />
```

```
</LinearLayout>
```

```
<LinearLayout
    android:id="@+id/tabLost"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center_horizontal"
    android:paddingStart="16dp"
    android:paddingEnd="16dp"
    android:clickable="true"
    android:focusable="true">
```

```
<TextView
    android:id="@+id/tvTabLost"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Lost"
    android:textColor="#1E293B"
    android:textSize="18sp"
    android:textStyle="bold" />
```

```
<View
    android:id="@+id/indicatorLost"
    android:layout_width="32dp"
    android:layout_height="3dp"
    android:layout_marginTop="6dp"
    android:background="@drawable/bg_tab_indicator"
    android:visibility="visible" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
<ListView
```

```

        android:id="@+id/lvLostItems"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:divider="@android:color/transparent"
        android:dividerHeight="14dp"
        android:clipToPadding="false"
        android:paddingBottom="80dp"
        android:scrollbars="vertical"
        android:nestedScrollingEnabled="true"
        android:overScrollMode="always" />

```

```

<TextView
    android:id="@+id/tvEmptyLost"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:gravity="center"
    android:text="No lost items posted yet.\nBe the first to report one!"
    android:textColor="#8E8A82"
    android:textSize="16sp"
    android:lineSpacingMultiplier="1.4"
    android:visibility="gone" />

```

```

</LinearLayout>

```

```

<ImageButton
    android:id="@+id/btnAddLost"
    android:layout_width="56dp"
    android:layout_height="56dp"
    android:layout_gravity="bottom|center_horizontal"
    android:layout_marginBottom="24dp"
    android:background="@drawable/bg_circle_button"
    android:src="@drawable/ic_add"
    android:elevation="6dp"
    android:contentDescription="Add Lost Item" />

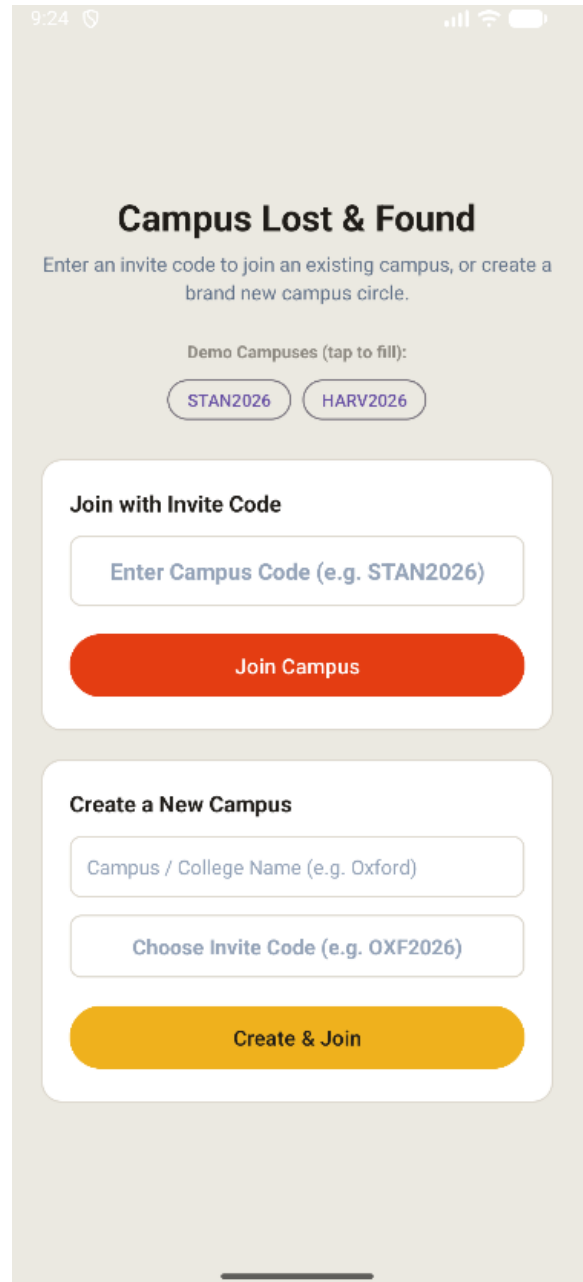
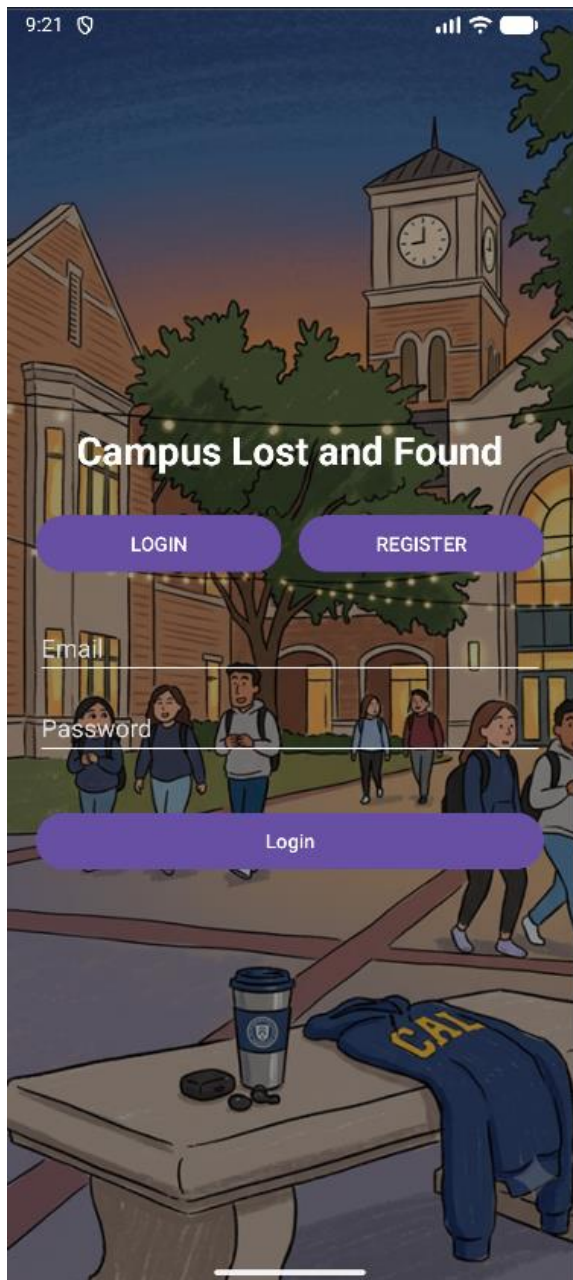
```

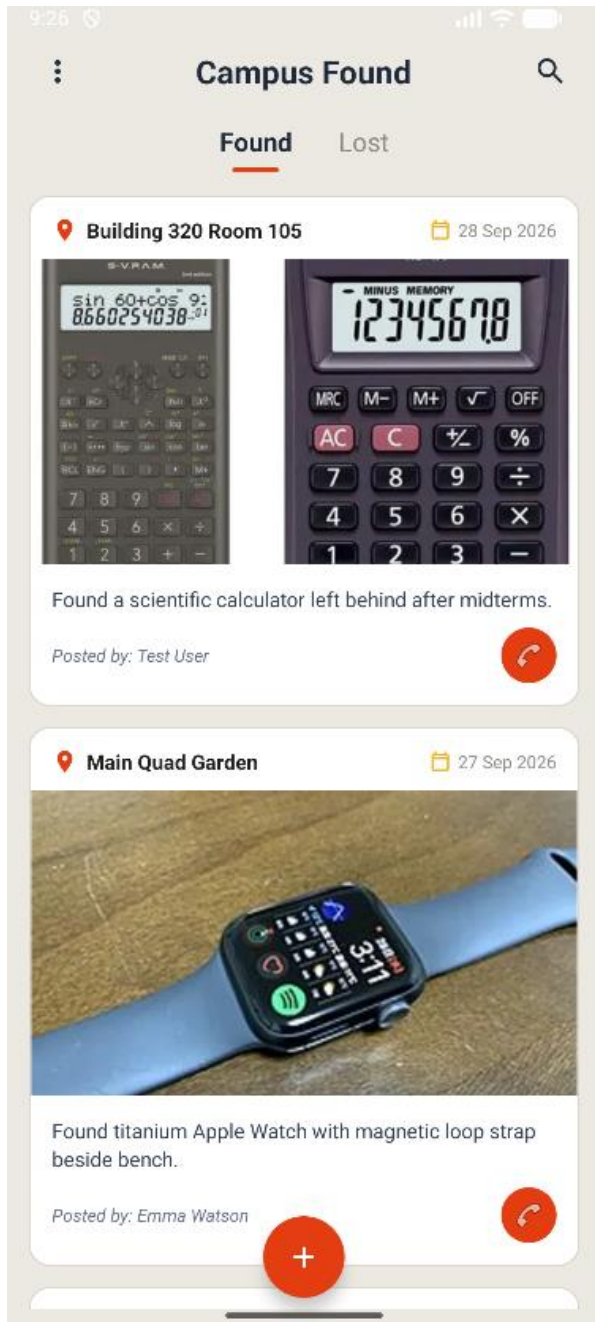
```

</FrameLayout>

```

OUTPUT





The screenshot shows the 'Report Found Item' form. At the top, there's a back arrow and the title 'Report Found Item'. The form includes the following sections:

- Item Photo:** A placeholder image showing a green and brown abstract pattern. Below it are two orange buttons: 'Camera' and 'Gallery'.
- Your Name (Optional):** A text input field containing 'Test User'.
- Your Email (Optional):** A text input field containing 'test@test.com'.
- Description *:** A text input field with the placeholder text: 'Describe the item (e.g. Black Lenovo wireless mouse with USB receiver)'.
- Location Found *:** A text input field with the placeholder text: 'Where was it found? (e.g. Central Library, 2nd Floor)'.
- Date Found *:** A text input field containing '30 Sep 2026'.
- Contact Phone Number *:** A text input field.

RESULTS AND DISCUSSION

The application successfully provides:

1. User registration and login.
2. Persistent user sessions.
3. Campus-based item isolation.
4. Separate Lost and Found feeds.
5. Item reporting with photographs.
6. SQLite-based local data storage.
7. Search and filtering functionality.
8. Custom ListView item cards using an Adapter.
9. Direct contact through the phone dialer.
10. Management and resolution of uploaded items.

APPLICATIONS

The Campus Lost & Found application can be used in:

1. Engineering colleges.
2. Universities.
3. Schools and educational institutions.
4. Corporate campuses.
5. Hostels and student residences.
6. Libraries and laboratories.
7. Sports complexes.
8. Large institutional campuses.
9. Events and conferences.
10. Training institutes

CONCLUSION

The **Campus Lost & Found** microproject successfully demonstrates the development of a Native Android application for managing lost and found items within campus communities.

The application provides important features such as user authentication, campus selection, Lost and Found feeds, item reporting, image selection, search, direct contact and item resolution.

The project also demonstrates practical implementation of Android development concepts including **Java, XML layouts, Activities, Intents, ListView, custom Adapter, SQLite database, Cursor, transactions, SharedPreferences and runtime permissions**.

The application provides a simple and organized approach to managing lost and found items and can be further enhanced by integrating a cloud database, real-time synchronization, push notifications and stronger authentication mechanisms.

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